Unit: Unit 4: Absolute Value
Topic: Definition of Absolute Value

Name: ____________
Date: ________

Advanced (Habanero)

Q1. What is the absolute value of -5?
(A) -5
(B) 5
(C) 0
(D) 10
(E) 25

Q2. The absolute value of a number x is defined as:
(A) x^2
(B) $sqrtx$
(C) $|x|$
(D) $x - 0$
(E) the distance from x to 0 on the number line

Q3. Evaluate $|2 - 3|$:
(A) -1
(B) 1
(C) 5
(D) 0
(E) 2

Q4. What is $|-2| + |3|$ equal to?
(A) 1
(B) 3
(C) 5
(D) 0
(E) 2

Q5. The absolute value of 0 is:
(A) 0
(B) 1
(C) -1
(D) 10
(E) 100

Q6. If $|x| = 5$, then x could be:
(A) 0
(B) 5
(C) -5
(D) 10
(E) -10

Q7. If $|x| = 5$, then x could be:
(A) 0
(B) 5
(C) -5
(D) -10
(E) 10

Q8. Which of the following is equal to $|x| + |y|$ when $x = -2$ and $y = 3$?
(A) 1
(B) 5
(C) -1
(D) 0
(E) 2

Open Ended Questions (FRQ)

Q9. Tom's house is 5 miles from the city center. If his friend's house is 3 miles from the city center in the opposite direction, what is the absolute value of the difference between the distances of their houses from the city center?

Q10. A company has a warehouse that is 12 miles from the main office. A new warehouse is 8 miles from the main office in the same direction. What is the absolute value of the difference between the distances of the two warehouses from the main office?

Chef's Tips

- Use number lines to visualize absolute value.
- Emphasize that absolute value is always non-negative.
- Make sure students understand that absolute value represents distance from zero.